

Using Propensity Scores

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Propensity scores

Matching

Weighting

Stratification

Direct Adjustment

...



Ingredients

150g unsalted butter, plus extra for greasing
150g plain chocolate, broken into pieces
150g plain flour
½ tsp baking powder
½ tsp bicarbonate of soda
200g light muscovado sugar
2 large eggs

Method

1. Heat the oven to 160C/140C fan/gas 3. Grease and base line a 1 litre heatproof glass pudding basin and a 450g loaf tin with baking parchment.
2. Put the butter and chocolate into a saucepan and melt over a low heat, stirring. When the chocolate has all melted remove from the heat.



estimand

estimator

estimate

Propensity scores

Matching

Weighting

Stratification

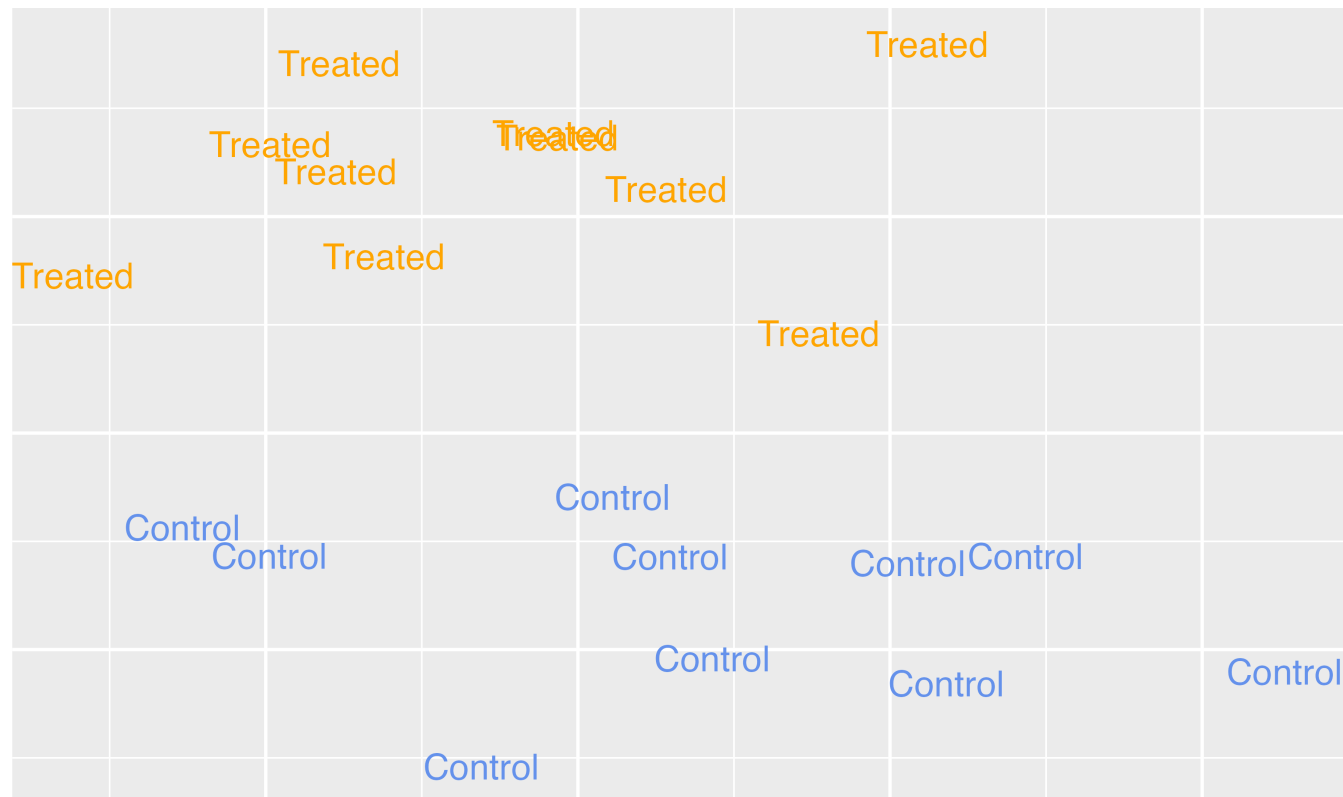
Direct Adjustment

...

Target estimands

Average Treatment Effect (ATE)

$$\tau = E[Y(1) - Y(0)]$$



Target estimands

Estimand	Target population	Example Research Question
ATE	Full population	<i>Should we decide whether to have extra magic hours all mornings to change the posted wait time for Seven Dwarfs Mine Train between 9-10 AM?</i> <i>Should a specific policy be applied to all eligible observations?</i>

Target estimands

Average Treatment Effect among the Treated (ATT)

$$\tau = E[Y(1) - Y(0) | Z = 1]$$

Target estimands

Estimand	Target population	Example Research Question
ATT	Exposed (treated) observations	<i>Should we stop extra magic hours to change the posted wait time for Seven Dwarfs Mine Train between 9-10 AM?</i> <i>Should we stop our marketing campaign to those currently receiving it?</i> <i>Should medical providers stop recommending treatment for those currently receiving it?</i>

Matching in R (ATT)

```
1 library(MatchIt)
2 m <- matchit(
3   net_num ~ income + health + temperature,
4   data = net_data
5 )
6 m
```

A `matchit` object

- method: 1:1 nearest neighbor matching without replacement
- distance: Propensity score
 - estimated with logistic regression
- number of obs.: 1752 (original), 908 (matched)
- target estimand: ATT
- covariates: income, health, temperature

Matching in R (ATT)

```
1 matched_data <- get_matches(m, id = "i")
2 as_tibble(matched_data)
```

```
# A tibble: 908 × 14
```

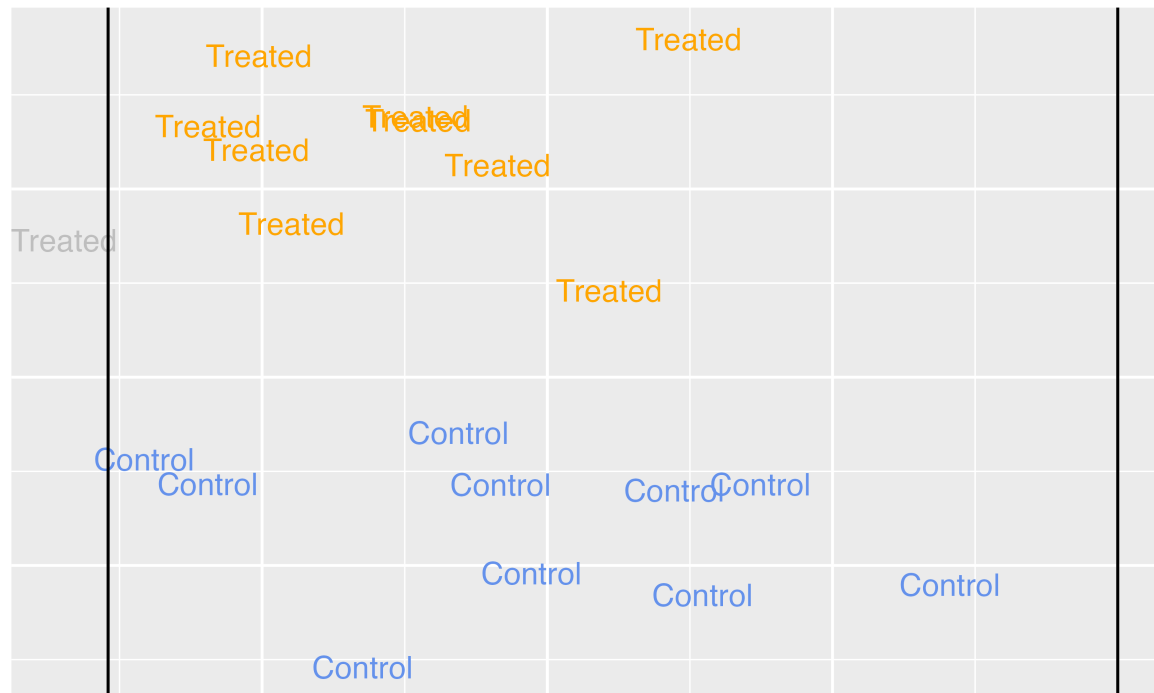
	i	subclass	weights	id	net	net_num	malaria_risk
	<chr>	<fct>	<dbl>	<int>	<lgl>	<int>	<dbl>
1	16	1	1	16	TRUE	1	15
2	109	1	1	109	FALSE	0	22
3	18	2	1	18	TRUE	1	28
4	1652	2	1	1652	FALSE	0	39
5	19	3	1	19	TRUE	1	18
6	64	3	1	64	FALSE	0	22
7	22	4	1	22	TRUE	1	17
8	329	4	1	329	FALSE	0	22
9	28	5	1	28	TRUE	1	22
10	706	5	1	706	FALSE	0	25

```
" - 888 - - - - -
```

Target estimands

Average Treatment Effect among the Unexposed / Controls (ATU / ATC)

$$\tau = E[Y(1) - Y(0) | Z = 0]$$



Target estimands

Estimand	Target population	Example Research Question
ATU	Unexposed (control) observations	<i>Should we add extra magic hours for all days to change the posted wait time for Seven Dwarfs Mine Train between 9-10 AM?</i> <i>Should we extend our marketing campaign to those not receiving it?</i> <i>Should medical providers extend treatment to those not currently receiving it?</i>

Matching in R (ATU)

```
1 m <- matchit(  
2   net_num ~ income + health + temperature,  
3   data = net_data,  
4   estimand = "ATC"  
5 )  
6 m
```

A `matchit` object

- method: 1:1 nearest neighbor matching without replacement
- distance: Propensity score
 - estimated with logistic regression
- number of obs.: 1752 (original), 908 (matched)
- target estimand: ATC
- covariates: income, health, temperature

`matchit()` spells this estimand **ATC**

Target estimands

Average Treatment Effect among the Matched (ATM)

Target estimands

Estimand	Target population	Example Research Question
ATM	Evenly matchable	<i>Are there some days we should change whether we are offering extra magic hours in order to change the posted wait time for Seven Dwarfs Mine Train between 9-10 AM?</i> <i>Is there an effect of the exposure for some observations?</i> <i>Should those at clinical equipoise receive treatment?</i>

Matching in R (ATM)

```
1 m <- matchit(  
2   net_num ~ income + health + temperature,  
3   data = net_data,  
4   link = "linear.logit",  
5   caliper = 0.1  
6 )  
7 m
```

The caliper is the largest difference between two observations we accept as a match, measured in standard deviations of the propensity score (here, on the linear logit scale). Observations that cannot be matched within it are discarded

Matching in R (ATM)

A `'matchit'` object

- method: 1:1 nearest neighbor matching without replacement
- distance: Propensity score [caliper]
 - estimated with logistic regression and linearized
- caliper: <distance> (0.036)
- number of obs.: 1752 (original), 898 (matched)
- target estimand: ATT
- covariates: income, health, temperature

Matching in R (ATM)

```
1 matched_data <- get_matches(m, id = "i")
2 as_tibble(matched_data)
```

```
# A tibble: 898 × 14
```

	i	subclass	weights	id	net	net_num	malaria_risk
	<chr>	<fct>	<dbl>	<int>	<lgl>	<int>	<dbl>
1	16	1	1	16	TRUE	1	15
2	109	1	1	109	FALSE	0	22
3	18	2	1	18	TRUE	1	28
4	1652	2	1	1652	FALSE	0	39
5	19	3	1	19	TRUE	1	18
6	657	3	1	657	FALSE	0	22
7	22	4	1	22	TRUE	1	17
8	329	4	1	329	FALSE	0	22
9	28	5	1	28	TRUE	1	22
10	706	5	1	706	FALSE	0	25

11. 0000 0000 0000 0000 0000 0000 0000 0000

Your Turn 1

Using the propensity scores you created in the previous exercise, create a “matched” data set using the ATM method with a caliper of 0.2.

Propensity scores

Matching

Weighting

Stratification

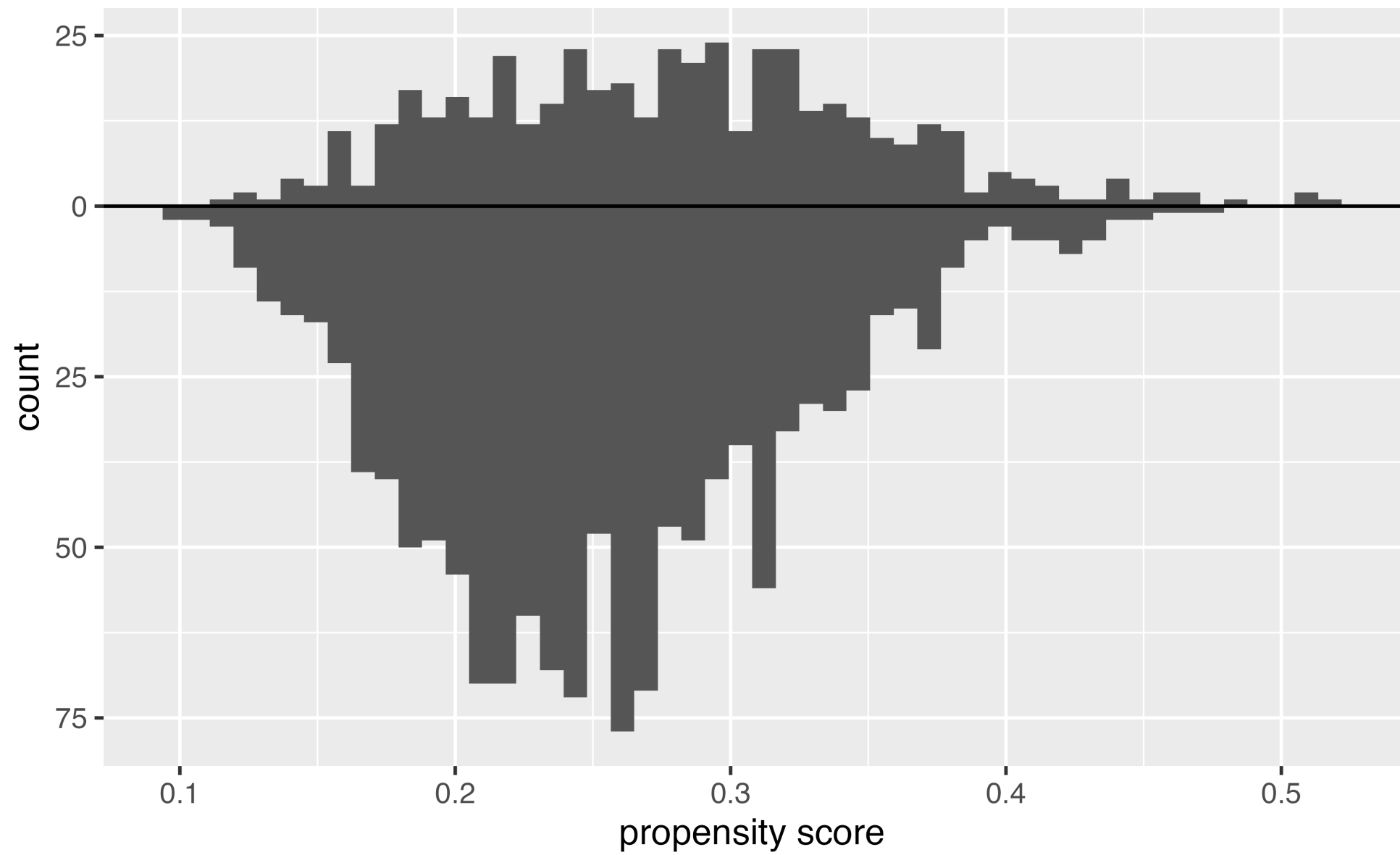
Direct Adjustment

...

Matching or weighting?

- Weighting is statistically more efficient; we recommend it when possible
- Matching has a distinct advantage: it's easy to understand

Histogram of propensity scores



Target estimands: ATE

Average Treatment Effect (ATE)

$$w_{ATE} = \frac{Z_i}{p_i} + \frac{1 - Z_i}{1 - p_i}$$

$$1 \cdot (Z / p) + ((1 - Z) / (1 - p))$$

Tilting functions

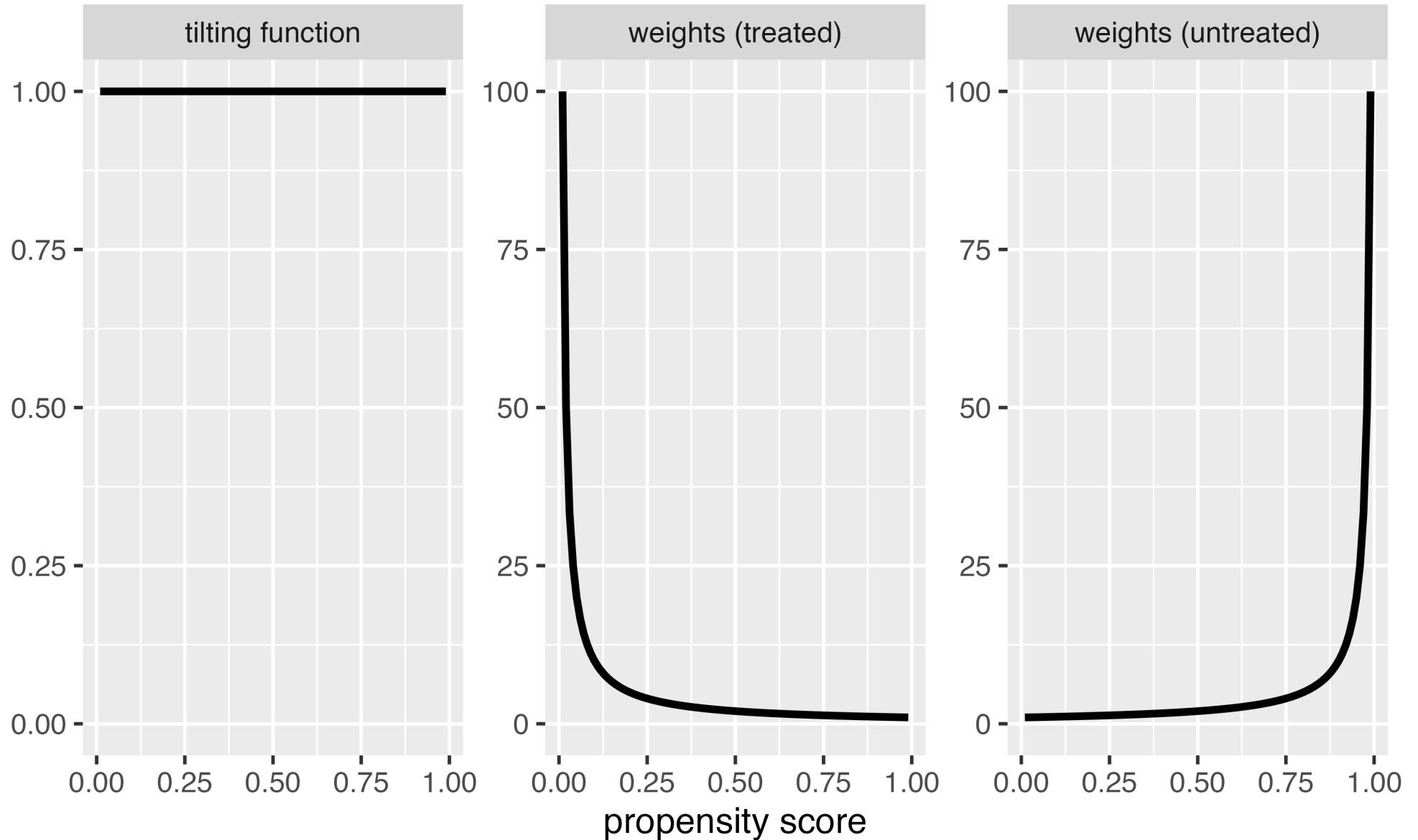
Every weight is the ATE weight times a **tilting function**, $h(p_i)$

$$w_i = \frac{h(p_i)Z_i}{p_i} + \frac{h(p_i)(1 - Z_i)}{1 - p_i}$$

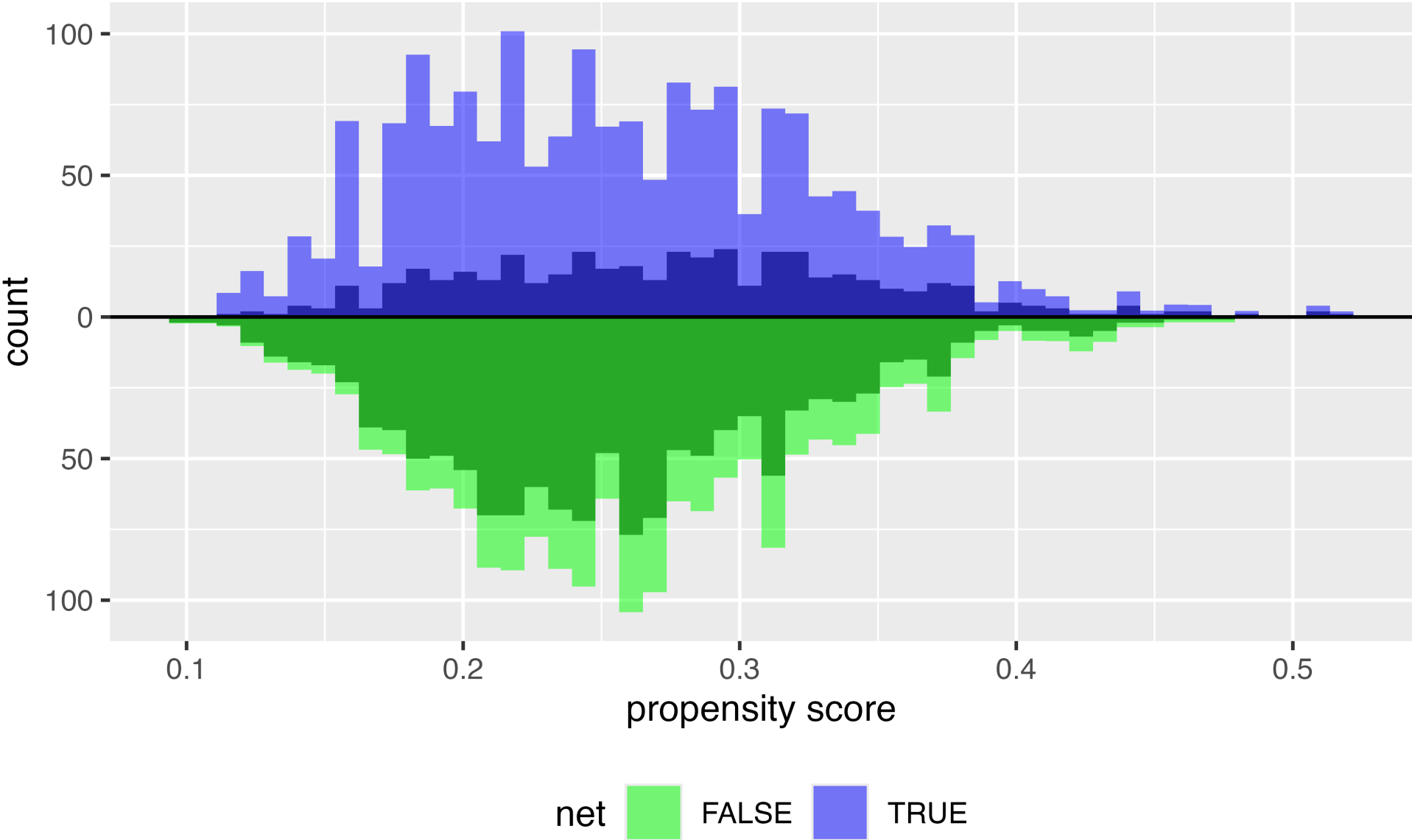
The tilting function reshapes the population we are targeting

The tilting function is $h(p_i) = 1$: the population is unchanged and the weights are unbounded

ATE tilting function



ATE



Target estimands: ATT & ATU

Average Treatment Effect Among the Treated (ATT)

$$w_{ATT} = \frac{p_i Z_i}{p_i} + \frac{p_i(1 - Z_i)}{1 - p_i}$$

$$1 \quad ((p * Z) / p) + ((p * (1 - Z)) / (1 - p))$$

The tilting function is $h(p_i) = p_i$: the treated get a weight of 1

Target estimands: ATT & ATU

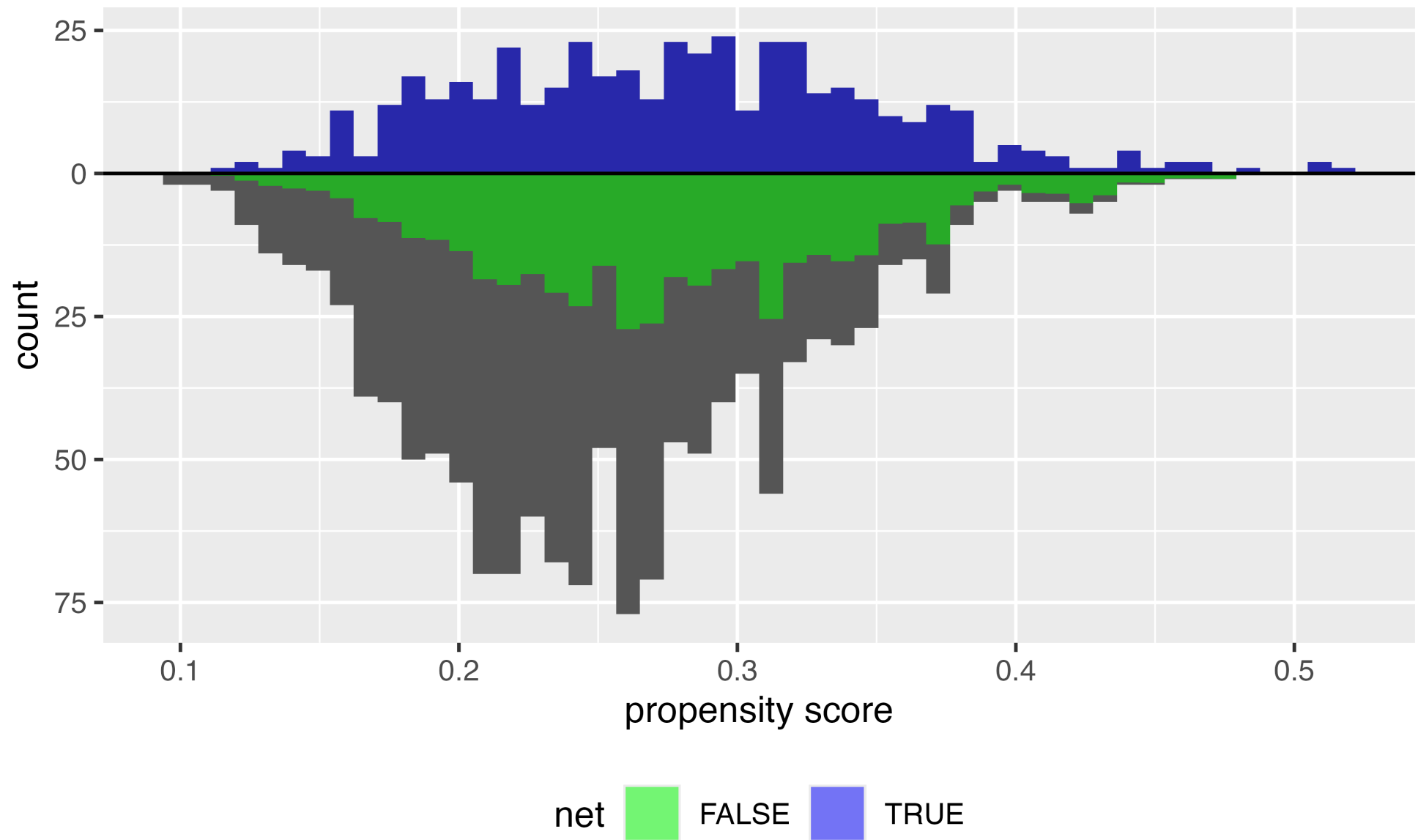
Average Treatment Effect Among the Unexposed (ATU)

$$w_{ATU} = \frac{(1 - p_i)Z_i}{p_i} + \frac{(1 - p_i)(1 - Z_i)}{(1 - p_i)}$$

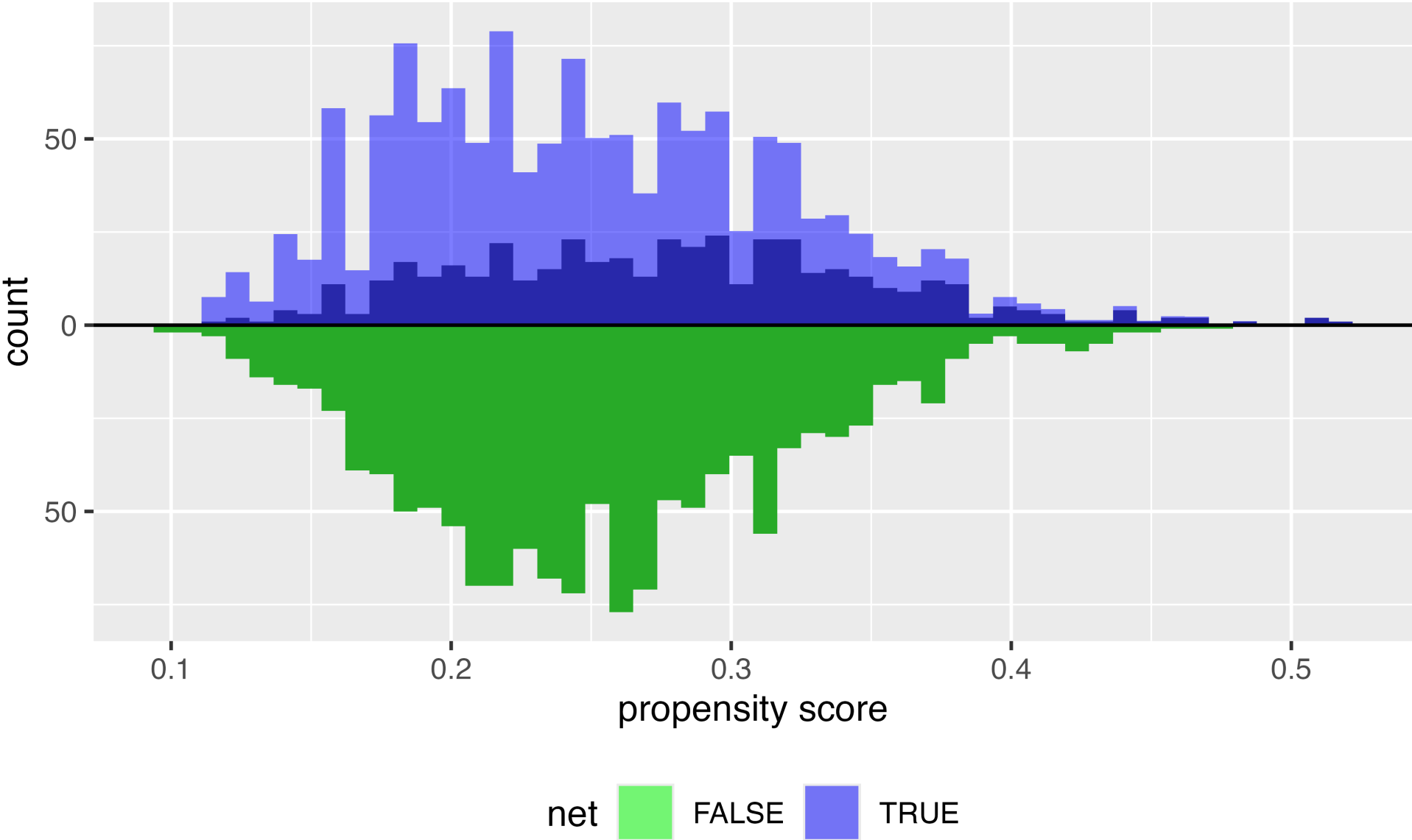
$$1 \left(\frac{((1 - p) * Z)}{p} + \frac{((1 - p) * (1 - Z))}{(1 - p)} \right)$$

The tilting function is $h(p_i) = 1 - p_i$: the untreated get a weight of 1

ATT



ATU



Target estimands: ATM & ATO

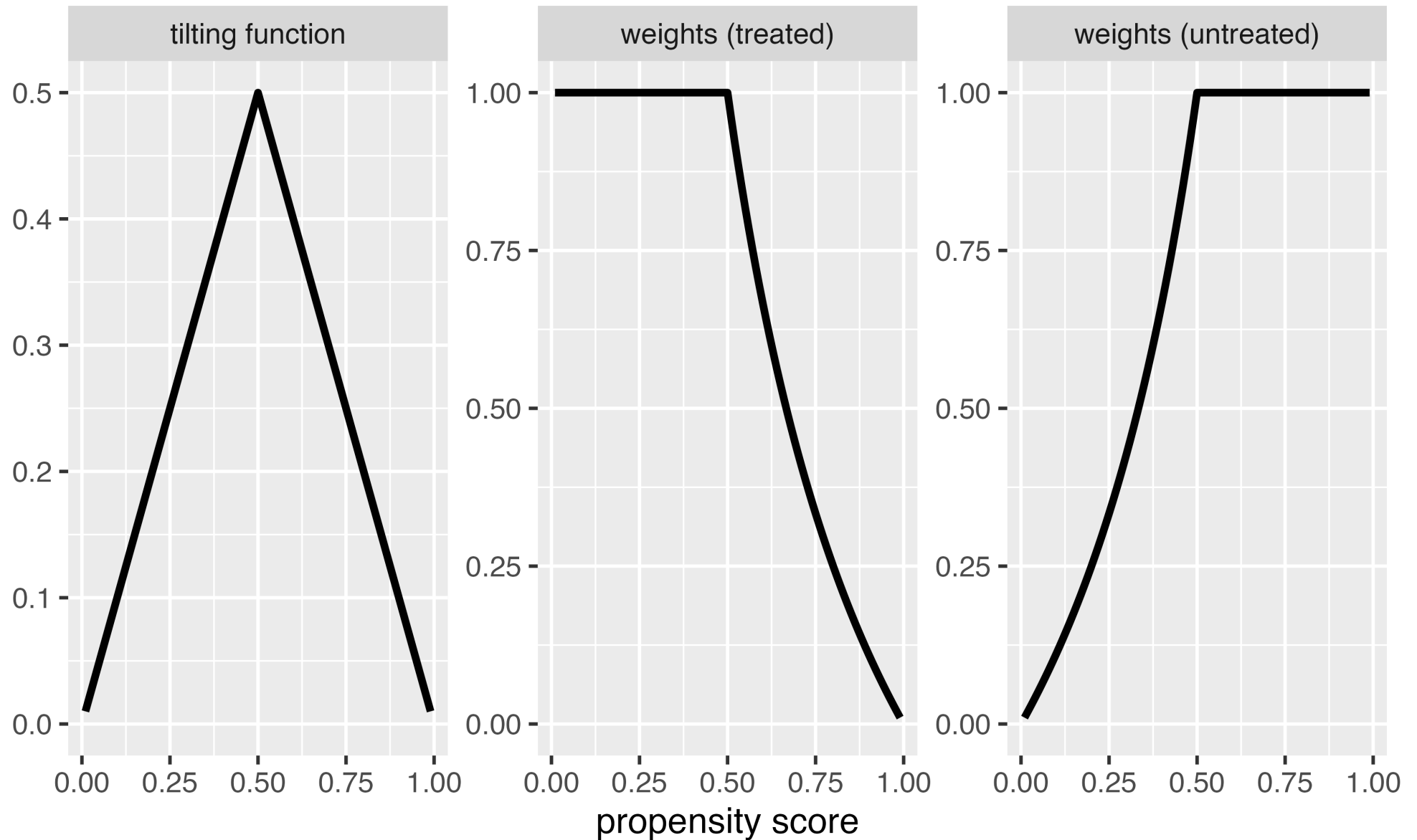
Average Treatment Effect Among the Evenly Matchable (ATM)

$$w_{ATM} = \frac{\min\{p_i, 1 - p_i\}}{Z_i p_i + (1 - Z_i)(1 - p_i)}$$

```
1 pmin(p, 1 - p) / (Z * p + (1 - Z) * (1 - p))
```

The tilting function is $h(p_i) = \min\{p_i, 1 - p_i\}$: a triangle peaking at 0.5

ATM tilting function



Target estimands: ATM & ATO

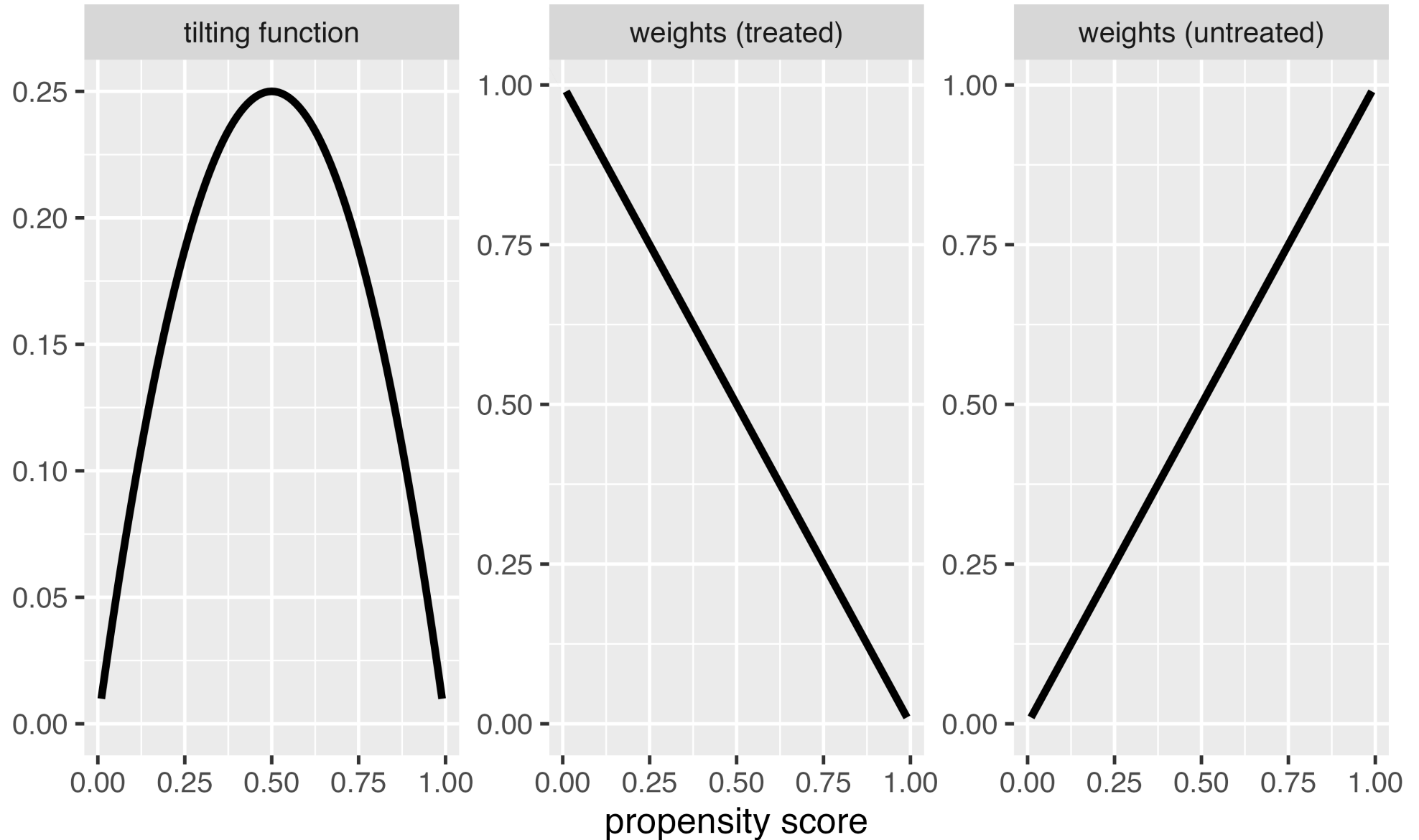
Average Treatment Effect Among the Overlap Population

$$w_{ATO} = (1 - p_i)Z_i + p_i(1 - Z_i)$$

$$1 \quad (1 - p) * Z + p * (1 - Z)$$

The tilting function is $h(p_i) = p_i(1 - p_i)$: smooth, peaking at 0.5, and the weights are bounded by 0 and 1

ATO tilting function

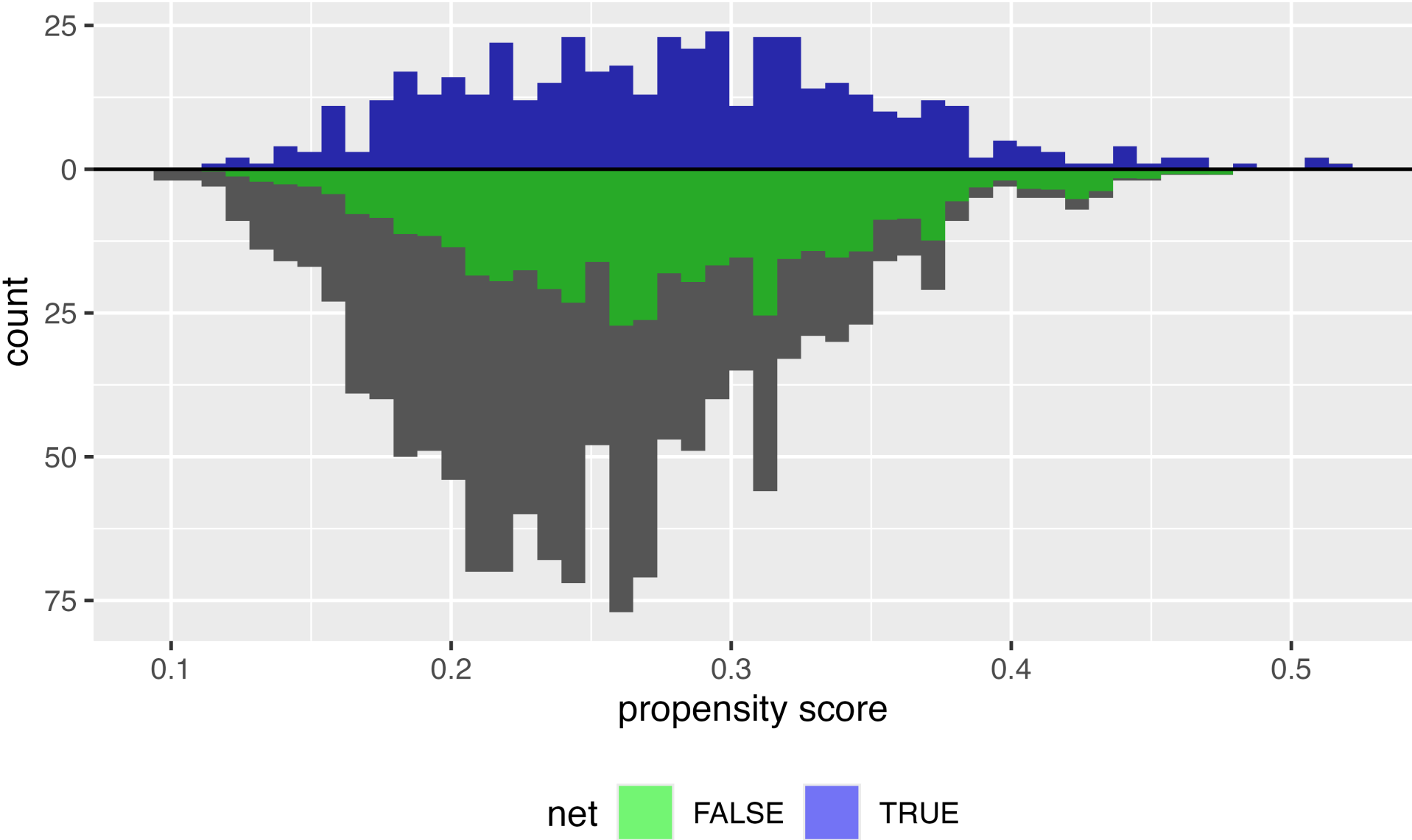


Target estimands

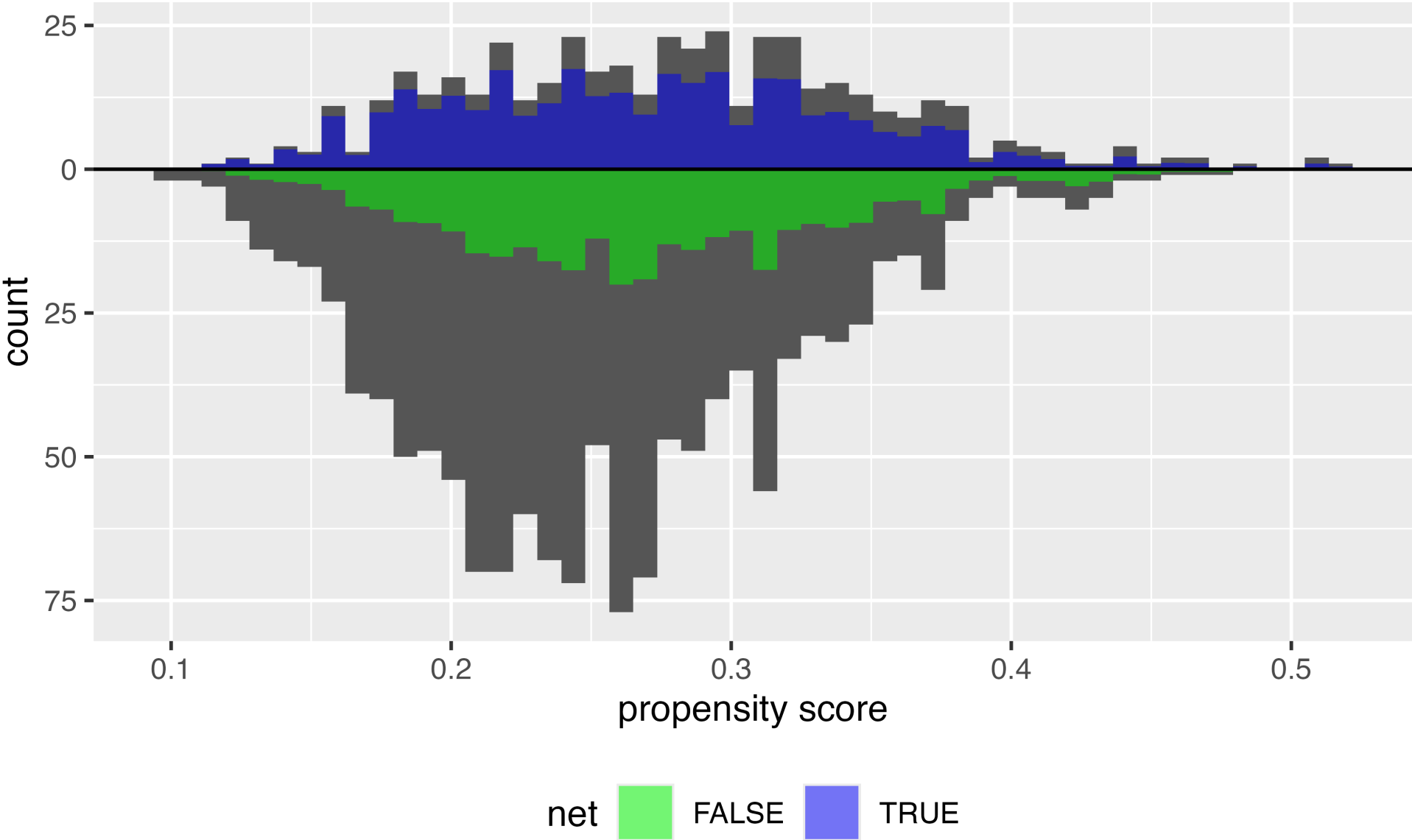
Estimand	Target population	Example Research Question
ATO	Overlap population	<i>Same as ATM</i>

We will compare these populations with the effective sample size when we check balance

ATM



ATO



ATE in R



Average Treatment Effect (ATE) $w_{ATE} = \frac{Z_i}{p_i} + \frac{1-Z_i}{1-p_i}$

```
1 library(propensity)
2 df <- propensity_model |>
3   augment(type.predict = "response", data = net_data) |>
4   mutate(w_ate = wt_ate(.fitted, net))
```

Extreme weights

- The minimum of the ATE weights is 1, but the maximum is unbounded
- The closer the probability of the observed exposure is to 0, the higher the weight
- **Extreme weights *destabilize*** our estimate, resulting in worse precision and potentially worse bias

Extreme weights

```
1 df |>
2   arrange(desc(w_ate)) |>
3   select(net, income, health, temperature, .fitted, w_ate)
```


Extreme weights

```
# A tibble: 1,752 × 6
```

	net	income	health	temperature	.fitted	w_ate
	<lgl>	<dbl>	<dbl>	<dbl>	<dbl>	<psw{ate}>
1	TRUE	501	10	29.9	0.117	8.556008
2	TRUE	456	7	26.4	0.120	8.328450
3	TRUE	544	10	29.7	0.126	7.915342
4	TRUE	424	13	20.2	0.136	7.361471
5	TRUE	613	43	30.4	0.138	7.270849
6	TRUE	644	33	31.9	0.139	7.190567
7	TRUE	532	26	24.6	0.143	7.011099
8	TRUE	502	21	22.7	0.143	6.968780
9	TRUE	633	35	29.7	0.145	6.883178
10	TRUE	547	18	24.7	0.146	6.845732

```
"# A tibble: 1,752 × 6"
```

Stabilized weights

```
1 df <- df |>
2   mutate(
3     sw_ate = wt_ate(.fitted, net, stabilize = TRUE)
4   )
5
6 df |>
7   summarize(mean = mean(sw_ate), min = min(sw_ate), max = max(sw_ate))
```

```
# A tibble: 1 × 3
  mean    min    max
<dbl> <dbl> <dbl>
1  1.00 0.501  2.22
```

Stabilizing puts the marginal probability of the observed exposure in the numerator: the proportion exposed for exposed units, the proportion unexposed for unexposed units

Trimming and truncation

- **Trimming:** set an acceptable range of the propensity score and drop the observations outside it
- **Truncation:** set the values outside the range to the minimum or maximum of the range
- Some authors use these terms interchangeably, or with the opposite meanings

Trimming and truncation in R

```
1 df <- df |>
2   mutate(
3     trimmed_ps = ps_trim(.fitted, method = "adaptive") |>
4       ps_refit(propensity_model),
5     trimmed = is_unit_trimmed(trimmed_ps),
6     trunc_ps = ps_trunc(.fitted, method = "pctl", lower = 0.01, upper = 1)
7   )
8
9 df |>
10  select(net, income, health, temperature, .fitted, trimmed) |>
11  filter(trimmed)
```

Trimming and truncation in R

```
# A tibble: 2 × 6
  net      income health temperature .fitted trimmed
<lgl>   <dbl>   <dbl>         <dbl>    <dbl> <lgl>
1 FALSE    340     15          26.3  0.0987 TRUE
2 FALSE    351     21          27.1  0.0980 TRUE
```

Trimming and truncation

- Trimming and truncation, like using a caliper, change the population we are making inferences about
- The tilting function is 0 for the observations we trimmed and 1 for the ones that remain
- You can think of the resulting estimand as another type of overlap population

Assessing positivity

- Trimming and truncation respond to extreme propensity scores without saying where they came from
- **positively** searches the covariates for the region responsible

Propensity scores with a violation

```
1 pos_model <- glm(  
2   exposure ~ x1 + x2 + region,  
3   family = binomial(),  
4   data = pos_violations  
5 )  
6  
7 pos_df <- pos_model |>  
8   augment(type.predict = "response", data = pos_violations) |>  
9   mutate(w_ate = wt_ate(.fitted, exposure))  
10  
11 pos_df |>  
12   summarize(min = min(.fitted), max = max(.fitted), max_wt = max(w_ate))
```

```
# A tibble: 1 × 3  
  min    max max_wt  
  <dbl> <dbl> <dbl>  
1 0.000107 1.000  473.
```


Which units did trimming drop?

```
1 pos_df |>
2   mutate(trimmed = is_unit_trimmed(ps_trim(.fitted, method = "adaptive"))) |>
3   group_by(structural = region == "b" & x2 > 1) |>
4   summarize(n = n(), trimmed = sum(trimmed))
```

A tibble: 2 × 3

	structural	n	trimmed
	<lgl>	<int>	<int>
1	FALSE	924	396
2	TRUE	76	38

Trimming dropped half of the 76 units that can never be exposed, and 396 units that can

Locating the region directly

```
1 check_port(pos_violations, exposure, c(x1, x2, region)) |>
2   tidy() |>
3   filter(low_support) |>
4   select(description, n, prevalence)
```

```
# A tibble: 3 × 3
```

	description	n	prevalence
	<chr>	<int>	<dbl>
1	x1<-0.5027	305	0.0459
2	x1>=0.9626 & x1<1.154	53	0.962
3	x2>=1.063 & region=b	69	0

Your Turn 2

Using the propensity scores you created in the previous exercise, add the ATE weights to your data frame

Stretch: Using the same propensity scores, create ATM weights